

FYLA BOREAL SERIES

User Manual

v04

2026

FYLA LASER S.L.

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WARNING

USE OF CONTROLS OR ADJUSTMENTS OR PERFORMANCE OF PROCEDURES
OTHER THAN THOSE SPECIFIED HEREIN MAY RESULT IN HAZARDOUS
RADIATION EXPOSURE.

IMPORTANT

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GENERAL INFORMATION

Introduction

All the instructions in this User Manual must be followed before installation and operation. Damage to persons, material or **BOREAL laser accessory** can be produced from not following the steps and indications of this Manual.

FYLA cannot be held responsible for any damage which results from using or working with the system described below. The BOREAL laser accessory must be only used by qualified personnel after reading this manual carefully.

Important Indicators

WARNING

CONTAINS SECURITY INSTRUCTIONS. NOT FOLLOWING THEM MAY RESULT IN IRREVERSIBLE DAMAGE.

IMPORTANT

Contains important information.

Warranty

FYLA LASER S.L., standard warranty guarantees its lasers to be free of defects for one year from the date of shipment but could be extended. Detailed information regarding the warranty for our products can be obtained through our sales team. Please consult with them the specific terms and conditions that apply to your purchase. This warranty is in lieu of all other guarantees expressed or implied and does not cover incidental or consequential loss. Damaged caused by the user to the laser/OEM laser and/or its accessories because of misuse (voluntary or accidental) of the equipment, will void the warranty.

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All the information and technology aspects described within this manual are covered under the patent EU international code PCT/ES2014/070382.

IMPORTANT _____

Save the shipping container and packing material for future shipping needs and keep the guarantee of your product.

BOREAL* Specifications

FYLA BOREAL SPECIFICATIONS**	
Spectral Range	Version from 400*** – 1000 nm ***Initial wavelength depends on the supercontinuum source
Output polarization	Unpolarized
Optical Output	Free space or Multimode fibre output (1 m) with FC/PC connector (FC/APC and Collimated output customizable)
Synchronization/Connections	IN – Coupling holder (Laser) IN – Power supply IN – USB port (System control)
Power Requirements	110V – 220V 50/60 Hz
Displayed Parameters (Controlled)	Min/Max wavelength selection Central wavelength/Bandwidth selection
Control modes	Remote (USB port)
Operating Temperature	20 – 30 °C
Storage Temperature	0 – 60 °C
Bandwidth	10 – 550 nm
Simultaneous electable bands	1
Resolution	1 nm
Insertion losses (full bandwidth)	≤ 10 % (free space output) ≤ 30 % (Multimode fibre output) ≤ 60 % (Single-mode fibre output)

* Laser accessory intended for integration into a host system or setup. The integrator is responsible for safety compliance.

**Specifications are subject to change without notice

BOREAL Identification

FYLA BOREAL SERIES IDENTIFICATION		
Series name	Version	Model
BOREAL	vx	Gap / F
Indicates the series of the laser	Indicates the version of the Boreal	Indicates if the model is Free space or Fiber output

Description of the FYLA BOREAL laser accessory

The acquired equipment, FYLA BOREAL Tunable Wavelength module, is supplied strictly as a laser accessory intended for integration into a host system or setup. A FYLA BOREAL module is a bandpass filter with tunable central wavelength and tunable bandwidth. It offers the possibility to select a laser's wavelength in a range from lasers starting wavelength to 1000 nm with a bandwidth from 10 nm to 550 nm. Consequently, it offers unparalleled performance for a wide range of applications that are not even possible with other types of light sources. Fig. 1 and Fig. 2 illustrate the basic proprieties of the FYLA BOREAL module.

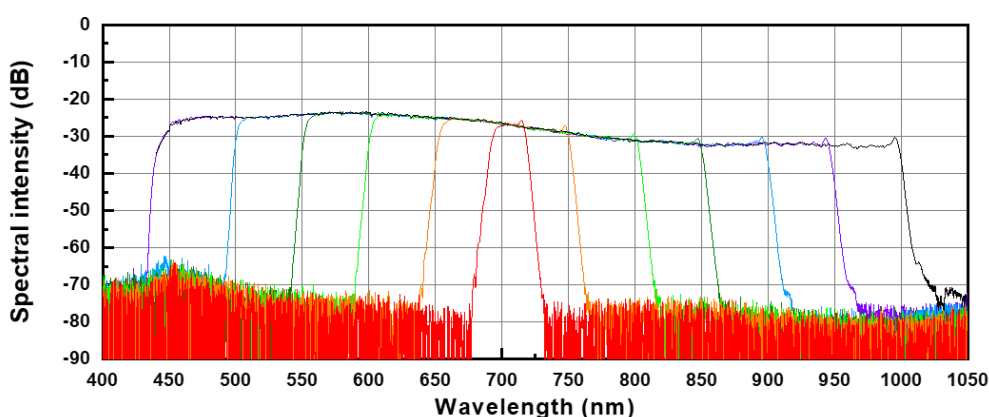


Fig 1. Supercontinuum laser emission filtered into different bands, maintaining the central wavelength while decreasing the bandwidth from 550 nm to 50 nm in 50 nm steps.

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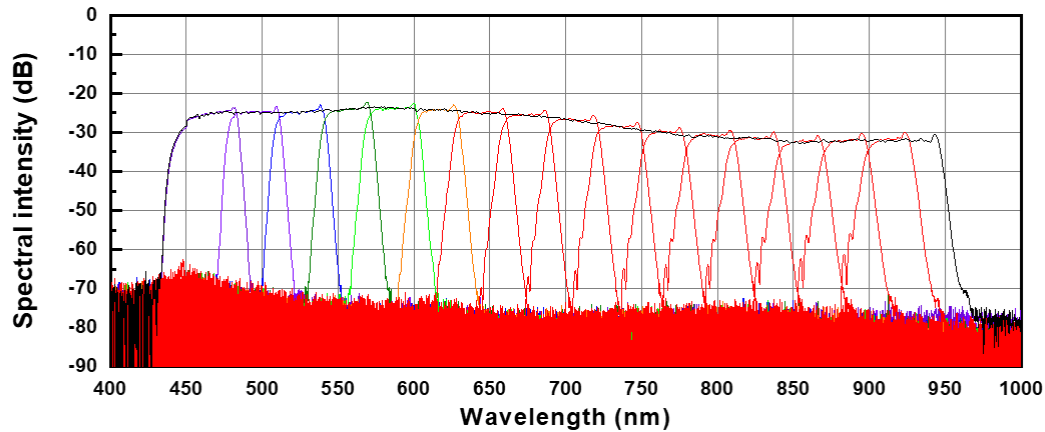


Fig 2. Supercontinuum laser emission filtered in different bands of 30 nm bandwidth and central wavelength progressively tuned with 30 nm steps.

USER SAFETY

Introduction

The FYLA BOREAL will always be used with FYLA's ultra-broadband laser sources. Your safe and effective use of this product is of utmost importance to us at FYLA. Please read the following laser safety information before attempting to operate the FYLA BOREAL combined with a supercontinuum source.

Laser Safety

The company FYLA LASER, S.L. has characterized the laser radiation at the output of the BOREAL laser accessory. Laser radiation may exceed Class 3B limits.

Therefore, BOREAL is sold exclusively as a laser accessory intended for integration into a laser system or setup.

As a laser accessory, BOREAL does not comply with IEC/EN 60825-1 safety requirements for end-user laser products and is not intended to be operated without complete safety architecture provided by the integrator.

The integrator is fully responsible for:

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- Implementing the safety controls required for the final laser class
- Preventing user access to hazardous radiation
- Providing compliant labels, user documentation, and safety instructions
- Performing the final laser classification of the host product.

FYLA is not responsible for injury, damage, or regulatory noncompliance resulting from operation of BOREAL laser accessory outside an integrated system with appropriate safety provisions.

As an accessory to the laser, a BOREAL filter can be used to select ranges of wavelengths within the visible spectrum. The use of said filter causes the laser output to have less power than the output of the source directly.

Even so and considering that the FYLA BOREAL can be easily removed, it is recommended to take the same protective measures when using it.

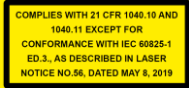
WARNING

**THE LASER RADIATION AT THE OUTPUT OF THIS UNIT MAY BE HARMFUL.
ALWAYS FOLLOW THESE PRECAUTIONS:**

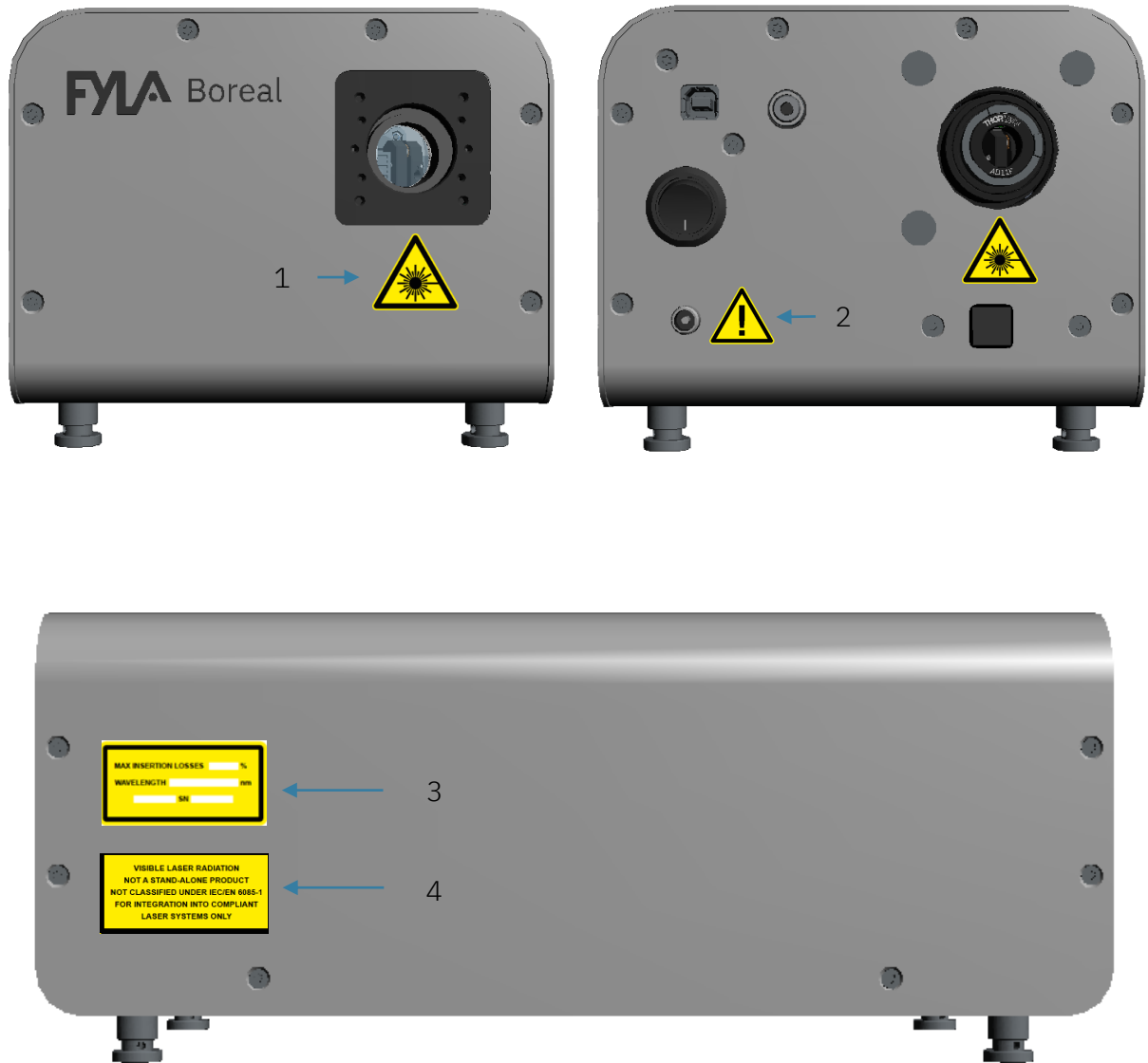
- ALWAYS WEAR PROTECTIVE GOGGLES OR EYEGLASSES APPROPRIATE FOR WORKING WITH CLASS 3B LASER LIGHT.
 - AVOID DIRECT EXPOSURE TO THE BEAM.
 - AVOID LOOKING AT THE BEAM DIRECTLY.
 - BE AWARE OF THE WARNINGS ON THE SAFETY LABELS STUCK ON THE EQUIPMENT.
 - THE LASER DISPOSES OF A HARDWARE INTERLOCK CONNECTOR WHICH CAN BE ACTIVATED IN CASE OF MACHINE FAILURE TO STOP THE EMISSION.
 - DO NOT OPEN THE LASER AND BOREAL SYSTEM. THERE ARE NO USER-SERVICEABLE PARTS INSIDE THE UNIT.
 - THE USER WILL NEVER NEED TO OPEN THE LASER SYSTEM. UNAUTHORISED OPENING OF THE LASER WILL VOID THE WARRANTY AND MAY RESULT IN UNDERPERFORMANCE OF THE LASER AND/OR IRREPARABLE DAMAGE TO THE INTERNAL COMPONENTS.
-

Labels and symbols identification

The following table explains the meaning of the different labels stuck to the BOREAL laser accessory and the reference number to identify them in the following pictures. Please be aware of them and use caution when working with the BOREAL laser accessory. Please use the same labels to properly indicate the area where the BOREAL is used.

Labels	Explanation	Number
	Radiation warning	1
	Caution, possible risk	2
	Boreal characteristics	3
	Explanation on laser accessory radiation	4
	Performance standard label	*

* Only used with equipment to be sent to EEUU.



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Laser Accessory Policy

The BOREAL laser module is supplied strictly as a laser accessory intended for integration into a host system or setup. It is not a finished laser product and therefore does not include the safety features required for a Class 3B laser product, such as:

- Key-controlled master switch
- Remote interlock connector
- Emission indicators
- Beam shutter or attenuator
- Protective housing interlocks

Because BOREAL is a component-level device, its final laser classification depends on the system into which it is integrated. The integrator or system manufacturer is responsible for:

1. Ensuring the complete system complies with IEC/EN 60825-1.
2. Providing all safety features required for the final class of the product.
3. Ensuring the user cannot access laser radiation above the class declared.
4. Applying appropriate labeling, documentation, and warnings.

BOREAL must not be used as a stand-alone device. It may only be operated inside a complete system or setup providing all mandatory safety mechanisms.

QUICK START

Unpacking the BOREAL laser accessory

Carefully unpack the tunable module system. Compare the contents against the list below and inspect them for any signs of damage. If parts are missing or you notice any signs of damage, please contact FYLA immediately.

IMPORTANT

Save the shipping container and packing material for future shipping needs and keeping the guarantee of your laser accessory.

WARNING

DO NOT OPEN THE LASER ACCESSORY. THERE ARE NO USER-SERVICEABLE PARTS INSIDE THE UNIT.

WARNING

THE USER WILL NEVER NEED TO OPEN THE BOREAL. UNAUTHORISED OPENING WILL VOID THE WARRANTY AND MAY RESULT IN UNDERPERFORMANCE OF THE LASER ACCESSORY AND/OR IRREPARABLE DAMAGE TO THE INTERNAL COMPONENTS.

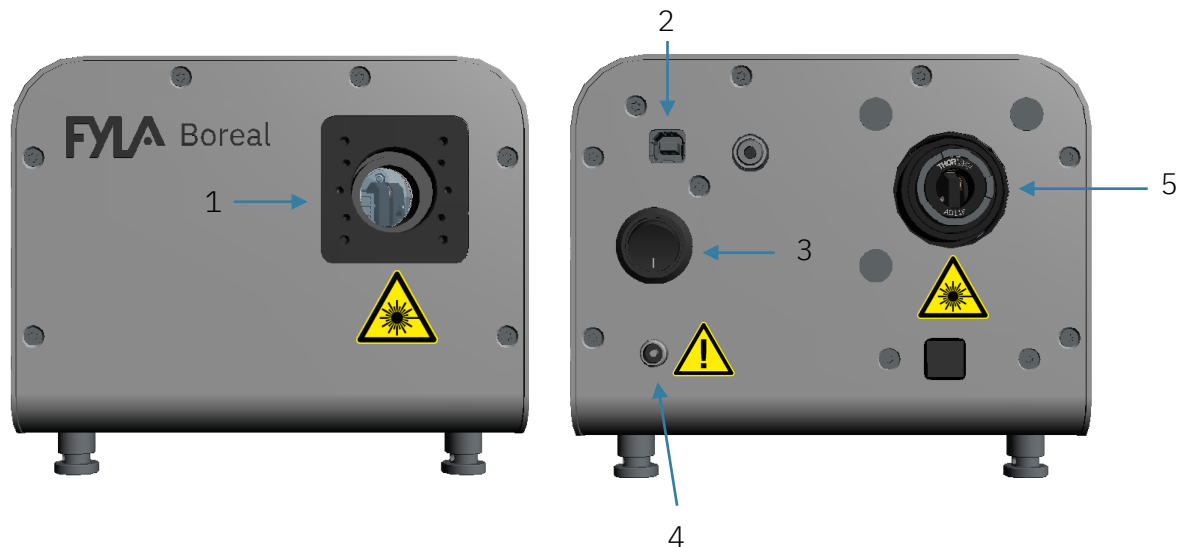
WARNING

IF THE WARRANTY STICKERS OF THE EQUIPMENT SHOW SIGNS OF HAVING BEEN REMOVED OR DAMAGED IN ANY WAY, THIS WILL VOID THE WARRANTY.

Items within a FYLA BOREAL series unit package:

- FYLA BOREAL UNIT
- AC Cable
- USB A-B Cable
- 24V Power Supply
- Specifications inspection sheet
- Fiber Output (in case you've purchased this output option)

System parts



1. BOREAL OUTPUT (COULD BE FREE SPACE OR FIBER OUTPUT)
2. USB PORT
3. POWER SWITCH ON/OFF
4. JACK CONNECTOR 24 V POWER SUPPLY SOCKET
5. BOREAL INPUT (LASER ENTRANCE)

Power supply

The BOREAL tunable module is configured at the factory for the line voltage and frequency appropriate for your country. If you are unsure how your unit is configured, check “AC Operating Voltages” for more details or ask FYLA through sales@fyala.com or support@fyala.com.

USING THE FYLA BOREAL LASER ACCESSORY

Electrical Connections

Switching ON

1. Place your FYLA BOREAL unit on your workbench, horizontally. Connect the AC cable to the 24V Power Supply. Then, connect the power's supply female JACK connector to the Rear Panel's male JACK connector.



Fig 1. 24V Power Supply JACK connector.

2. Turn ON the power switch of the rear panel to switch on the BOREAL module. Next, wait 30 seconds and the system will be getting ready for use.



Fig 2. Rear panel power switch

Switching OFF

To deactivate the BOREAL module, turn off the power switch from the rear panel and unplug the female JACK connector.



Fig 5. Rear panel power switch

Optical Connections

Optical Input

WARNING

THE LASER RADIATION EMITTED FROM THE SUPERCONTINUUM SOURCE MAY BE HARMFUL. PLEASE FOLLOW ALL THE SAFETY INSTRUCTIONS INDICATED IN THE SAFETY SECTION BEFORE OPERATING THE LASER + BOREAL SYSTEM.

MAKE SURE THAT THE LASER IS SWITCHED OFF WHEN CONNECTING IT TO THE BOREAL LASER ACCESSORY.

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IMPORTANT _____

The laser output must be connected to the BOREAL laser accessory input before starting the operation. Make sure that the laser is switched OFF before connecting it to the BOREAL.

IMPORTANT _____

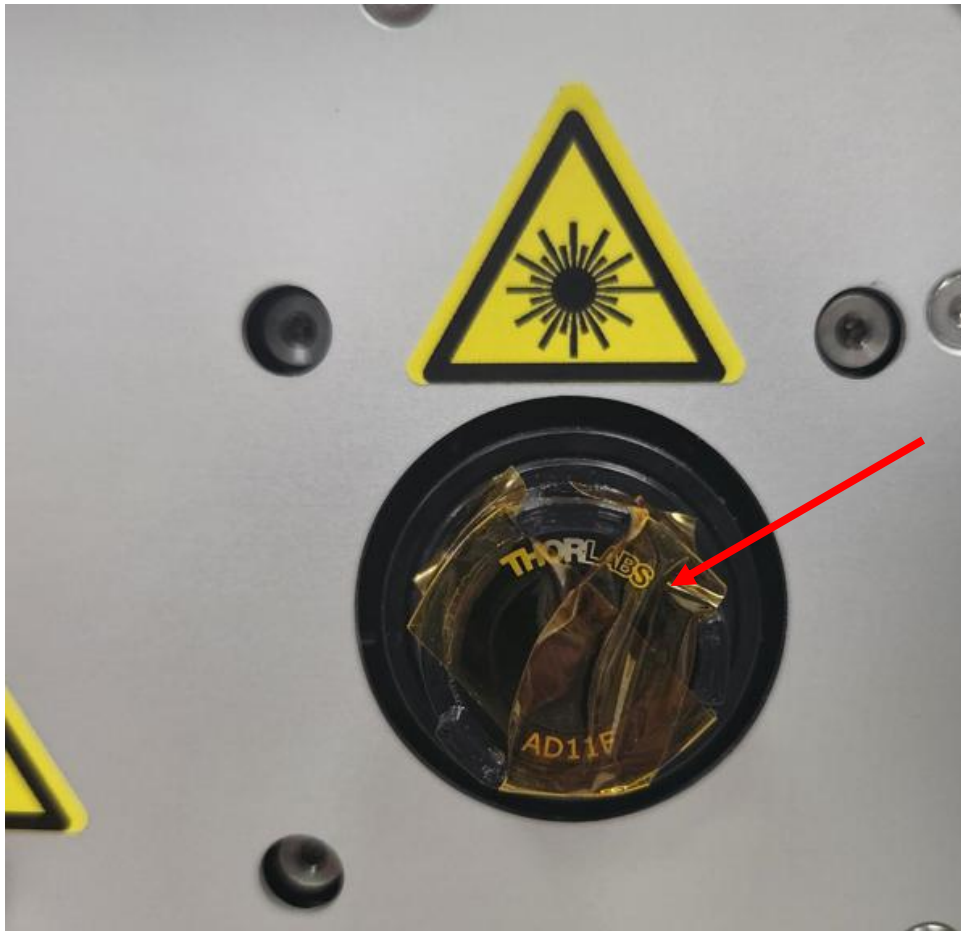
Follow the steps indicated below to achieve a correct BOREAL functionality.

It is very important to follow the next steps to achieve a correct connection of the laser to the BOREAL laser accessory. Please, follow the instructions of the corresponding section, depending on the laser you acquired.

- **ICEBLINK AND HORIZON SERIES:**

1. Remove the Laser's output protector and BOREAL's input protection tape. Please do not force the BOREAL's input connector when removing the tape.





2. The Laser's Output Collimator has a special mechanical part integrated, which matches the input of the BOREAL laser accessory. Insert the laser's output into the input and match both mechanical parts (Laser's Output & BOREAL's Input).



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Unmatched Connection



Matched Connection

3. Carefully tighten the screws from the BOREAL's input connector:



- **HORIZONV2 SERIES:**

1. If the collimator is currently attached to the fiber output of the HORIZONV2, unscrew the collimator from the fiber connector gently.



Use the provided fiber connector dust cap to avoid any damage to the surface of the output connector of the laser.



WARNING

WHEN CONNECTING OR DISCONNECTING THE FIBER OUTPUT CONNECTOR OF THE LASER TO THE COLLIMATOR, DO IT VERY CAREFULLY AND AVOID TOUCHING THE FIBER CONNECTOR TO ENSURE THAT IT IS ALWAYS CLEAN AND IN GOOD CONDITION.

2. Remove the 3D printed cover of the input mount of the BOREAL and the cover of the collimator. Please note that the input mount of the BOREAL is placed at the rear part of the equipment, where the switch on/off button is, as you can see in the following pictures:

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IMPORTANT

Do not take out of the collimator the ring glued to it, it must remain between the collimator and the BOREAL input mount.

3. Screw the collimator to the BOREAL's input mount until it is completely inserted into the mount thread. Turn the collimator clockwise until it stops. Avoid using excessive force during removal to prevent damage.



4. Once the collimator is connected to the BOREALs input, **insert and screw** the laser fiber output connector to the collimator.



Optical Output

Please follow the next steps to obtain light at the output:

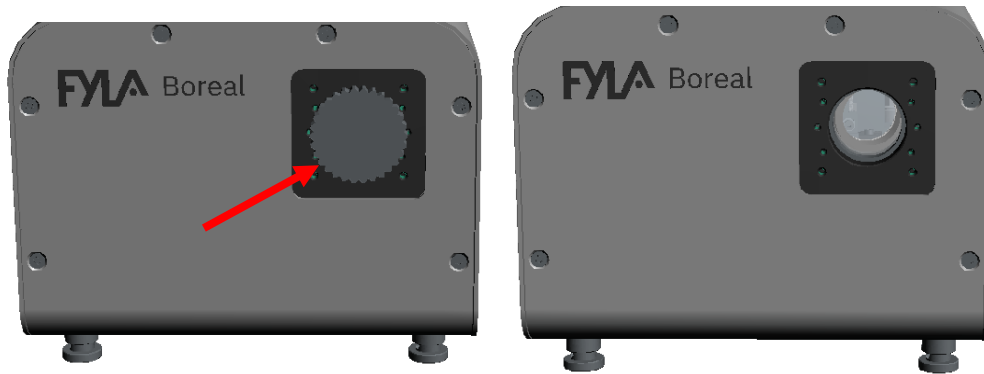
The BOREAL can have different output configurations:

- **Free Space Output**

Carefully remove the BOREAL's output protector and make sure that the output filter is clean

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- **Fiber output:**

Connect the fiber supplied by FYLA to the PAFA's output connector (picture below) and tighten it carefully.



- Switch from Fiber output to Free space BOREAL:

Given the case that you would want to use the BOREAL laser accessory as a free space element, you only need to take out the four screws that fix the PAFA. If you want to go back to Fiber output, fix the screws again and connect the fiber to the PAFA as previously explained.



Remote Control via User interface

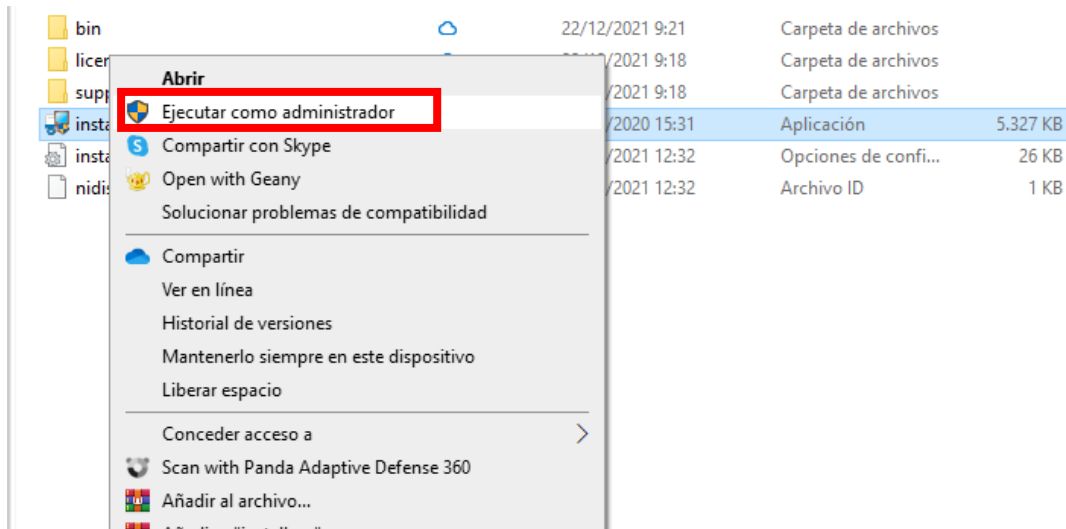
The FYLA BOREAL laser accessory must be controlled using the FYLA LASER User Interface (UI) software. All the functions can be controlled by the computer. The permanent communication between UI software and the BOREAL laser accessory provides you real-time subsystem information.

To achieve correct communication between software and laser please follow the next steps:

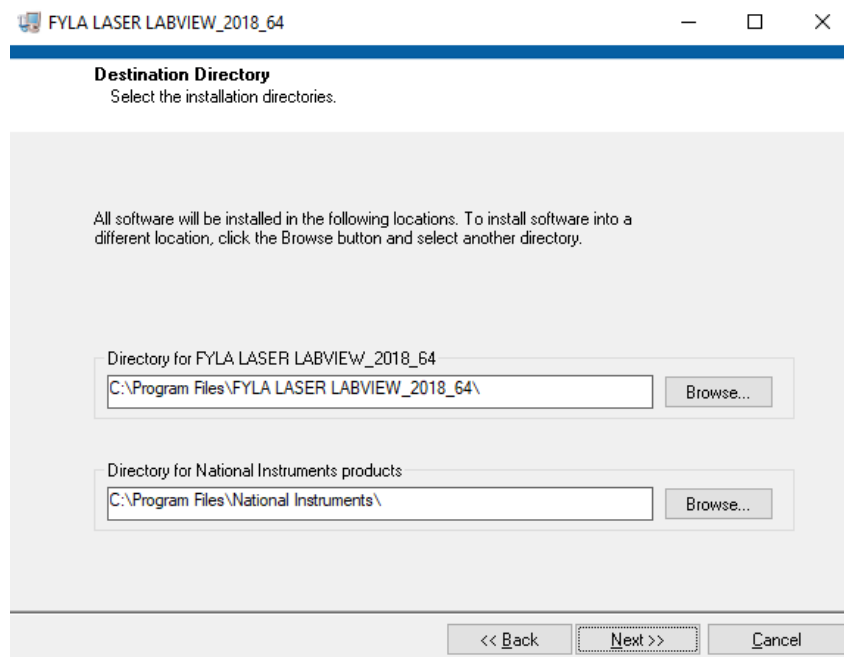
1. Start Windows 7 or higher version.
2. Open the attached USB sent by FYLA and copy the folders to your desktop.
3. Open the BOREAL user interface folder.
4. Next, open the “Installer” and “Volume” folders.
5. Click with left button one time over “install”

	bin		22/12/2021 9:21	Carpeta de archivos	
	license		22/12/2021 9:18	Carpeta de archivos	
	supportfiles		22/12/2021 9:18	Carpeta de archivos	
	install		05/11/2020 15:31	Aplicación	5.327 KB
	install		17/12/2021 12:32	Opciones de confi...	26 KB
	nidist.id		17/12/2021 12:32	Archivo ID	1 KB

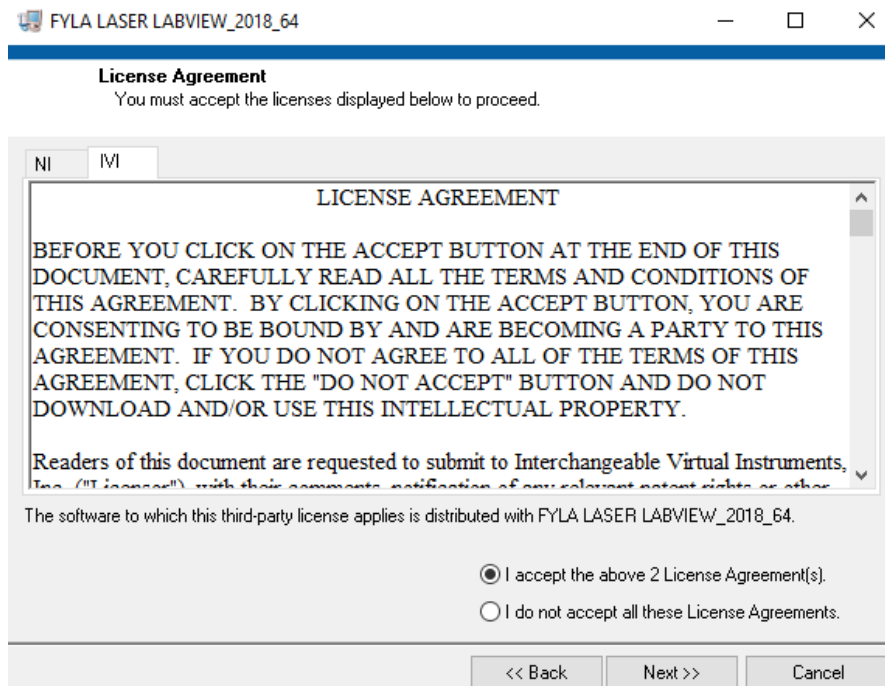
6. Click with right button one time over “install” and will appear the next photo. You will have to press the second option “Run as administrator” (red box).



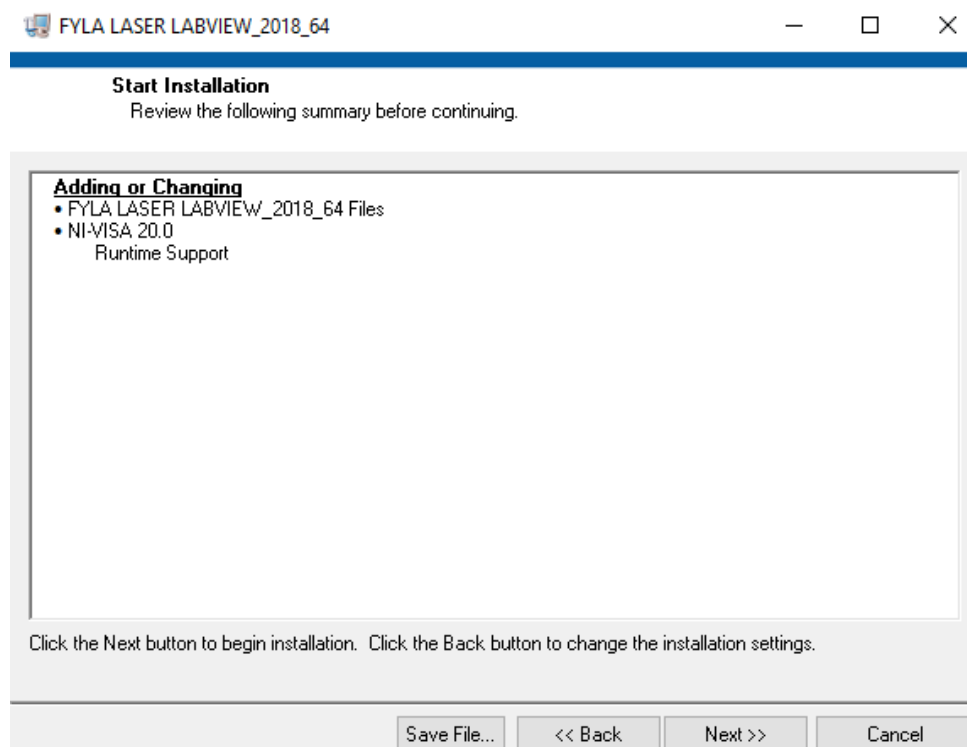
7. After that, will appear a window. You will have to press the left option (where it says: YES)
8. Then, select the destination directory and press “Next”



9. You will have to accept and press “Next”

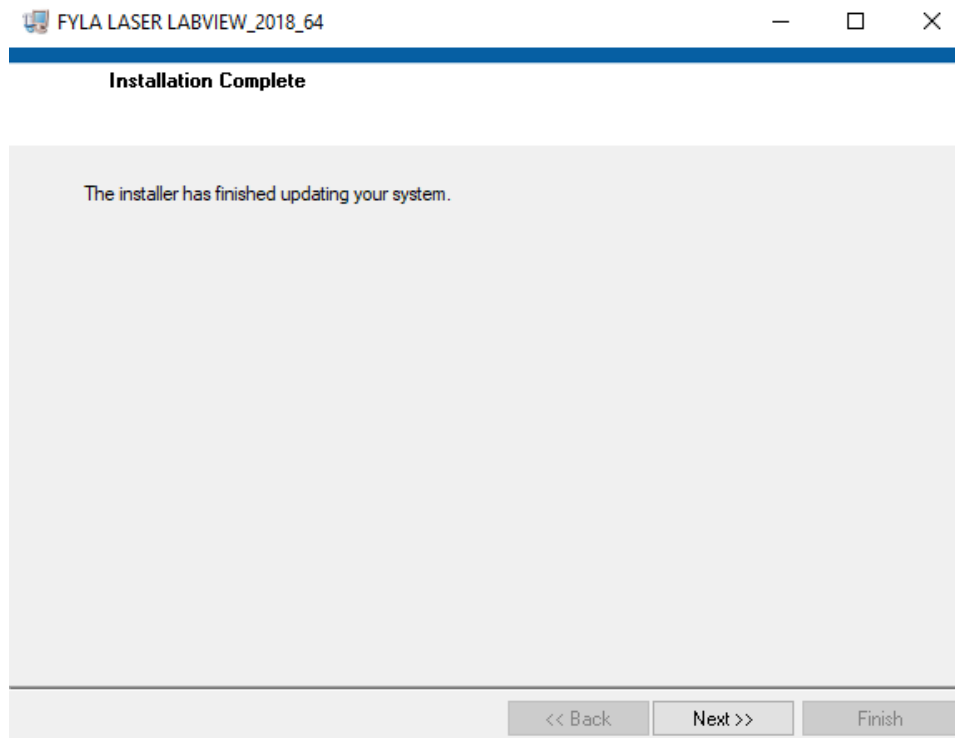


10. It will show the next photo and press “Next”

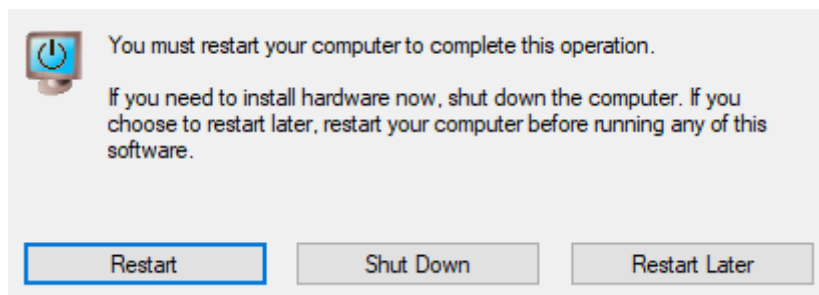


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11. Finally, press “Next”



12. Choose the action you want



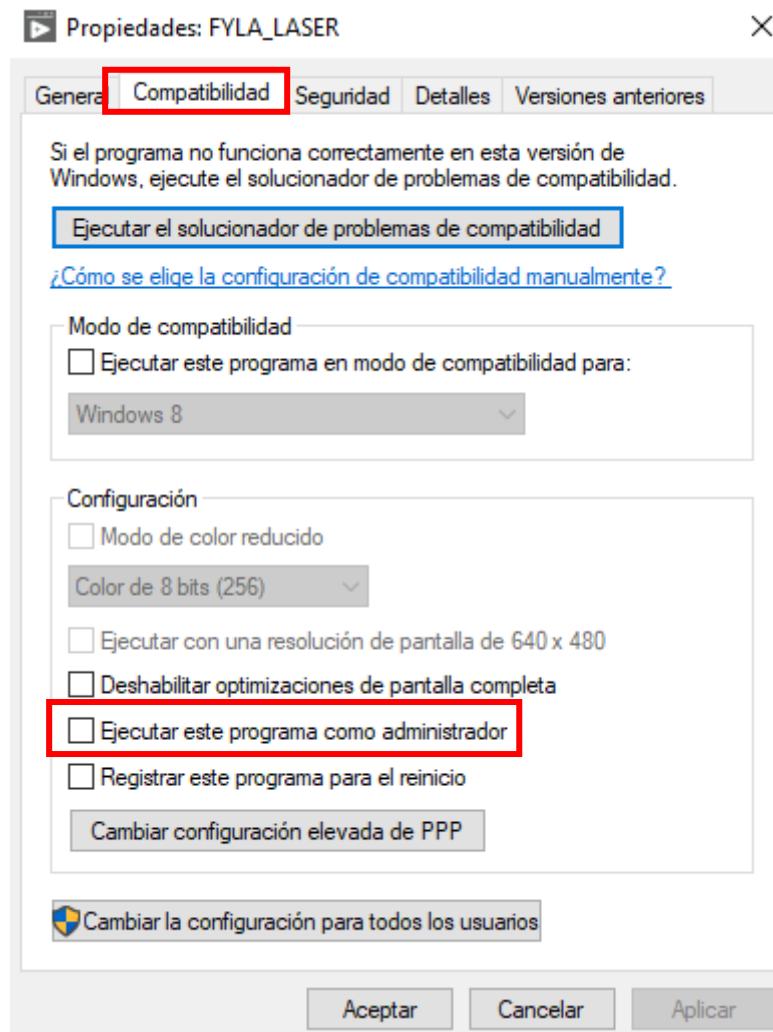
13. When the computer has been restarted, open again the first downloaded folder and enter in the “FYLA LASER” folder.

14. Finally, you will have to open “FYLA_LASER” and run as administrator.

Config	22/02/2022 13:09	Carpeta de archivos	
FYLA_LASER.aliases	17/12/2021 12:31	Archivo ALIASES	1 KB
FYLA_LASER	17/12/2021 12:31	Aplicación	3.191 KB
FYLA_LASER	17/12/2021 12:31	Opciones de confi...	1 KB

15. Press right click on the “FYLA_LASER” and select the last option “Properties”

16. It will show next window and select “Compatibility”. Then, click on “Run this program as an administrator”



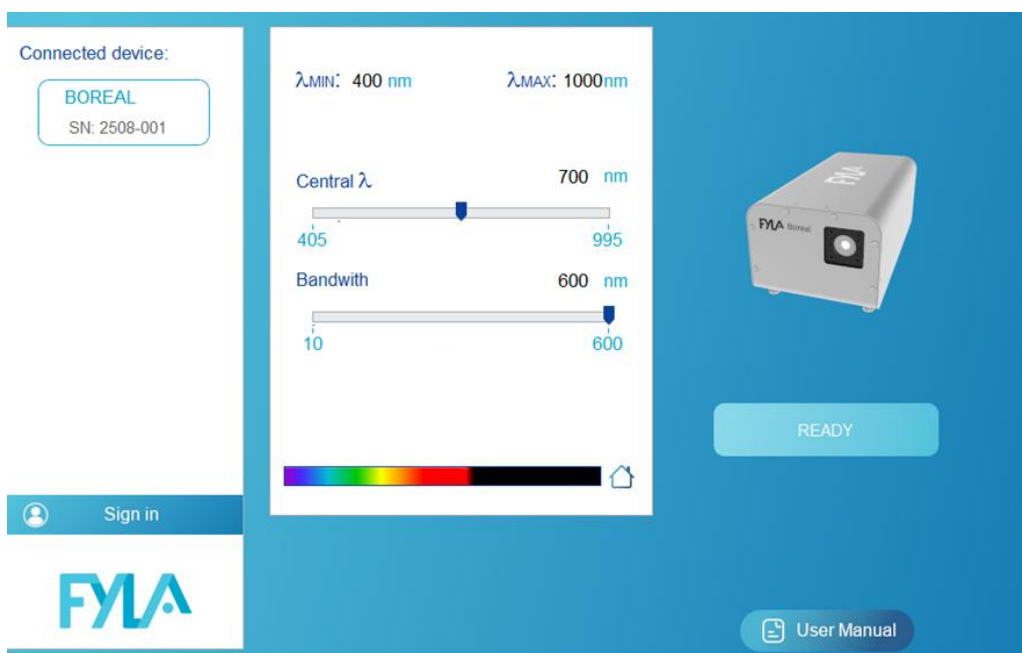
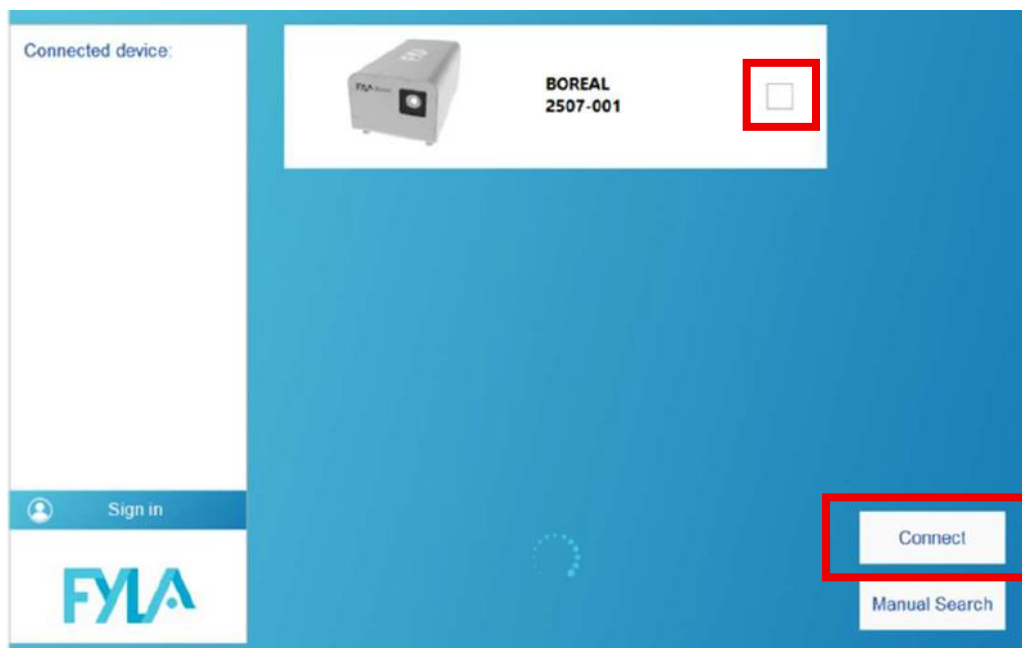
17. Connect the USB cable provided by FYLA to your PC and to the product. The USB port works in VCP (Virtual COM Port) mode. If your computer does not recognize the device automatically, please install the drivers from the next link:

<http://www.ftdichip.com/Drivers/VCP.htm>

18. Turn on the BOREAL laser accessory as previously explained, and the following screen will appear:



19. When the state of the BOREAL changes from “Moving” to the serial number corresponding to your product, first click the selector box and then the “Connect” button and the user screen will be shown (see next figures). In case you want to select COM port, use the “Manual Search” function.



At this point, the BOREAL is at its home position* and in READY state. When the laser is ON, light from $\lambda_{\min}$ to $\lambda_{\max}$ will be seen at the output. To change the band values, move to the next step.

*HOME position does not necessarily have to be that of the example picture.

20. To select a band, follow the next steps:

- First set min or max lambda values.
- Second, press Enter or click the blank space of the interface to send the command to the BOREAL.

The BOREAL laser accessory will start to move and will show the message “MOVING”. Once the band is set, the “READY” message will be shown.



The UI also allows to change the band values through the central wavelength / bandwidth setting. Write the desired values in the Central λ or Bandwidth fields, or move one of the markers to the position of those values.



WARNING

IN CASE THERE IS NO LIGHT AT THE OUTPUT WHEN THE SOFTWARE INDICATES THAT THE COLOUR SCALE IS ACTIVE, PLEASE TURN OFF IMMEDIATELY THE LASER AND CONTACT FYLA AT SALES@FYLA.COM / SUPPORT@FYLA.COM

Remote control via Serial Communication

The FYLA BOREAL laser accessory can be controlled remotely via serial communication (RS232), using the USB connector in the front Panel. Please follow the next steps to achieve the communication:

1. Connect the USB AB cable between the PC and the BOREAL after it is ready to be used.
2. Check the number of the COM port: Control panel / Device Manager

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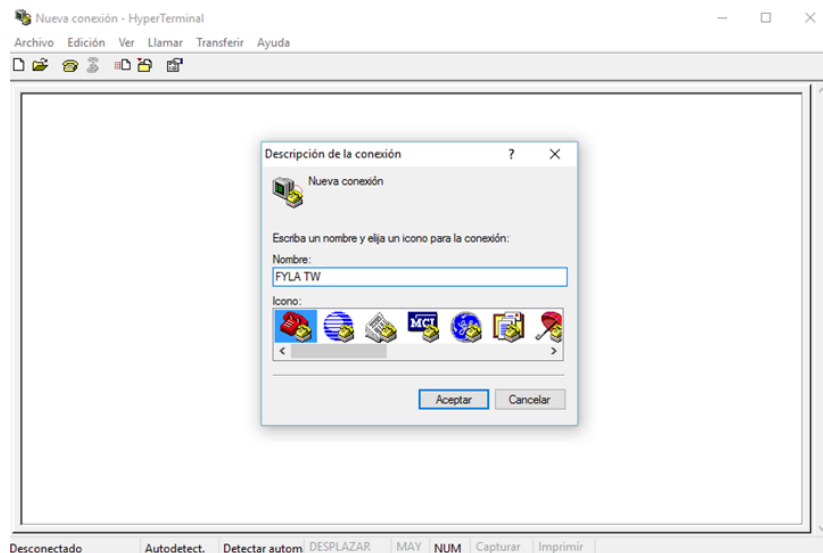
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- Open Windows HyperTerminal and create a new connection. The name of the connection in this example is “FYLA TW”, but you can choose another name.

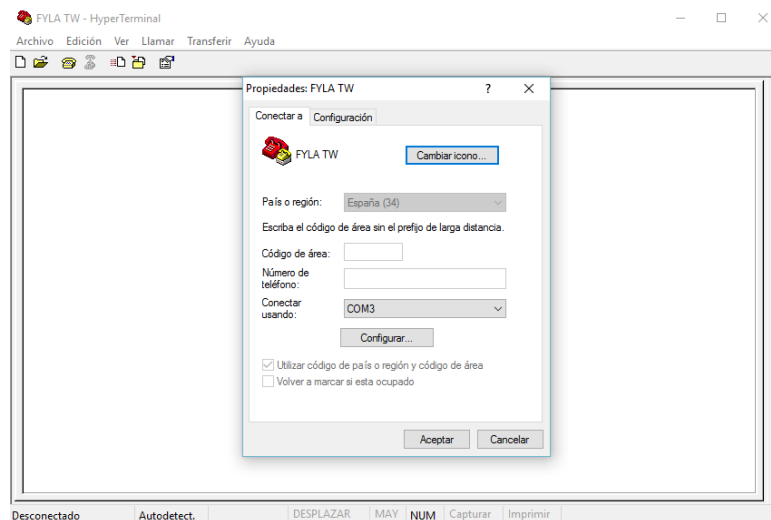
The USB port works in VCP (Virtual COM Port) mode. If your computer does not recognize the device automatically, you can install the driver from this link:

[VCP Drivers - FTDI \(ftdichip.com\)](http://ftdichip.com)

Communication can be performed with any program of Terminal for RS232. In this case, we are using Windows HyperTerminal.



- Select the detected COM port of the BOREAL.



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5. Next, introduce the port configuration and click Accept.

Bits por segundo: 19200

Bits de datos: 8

Paridad: Ninguno

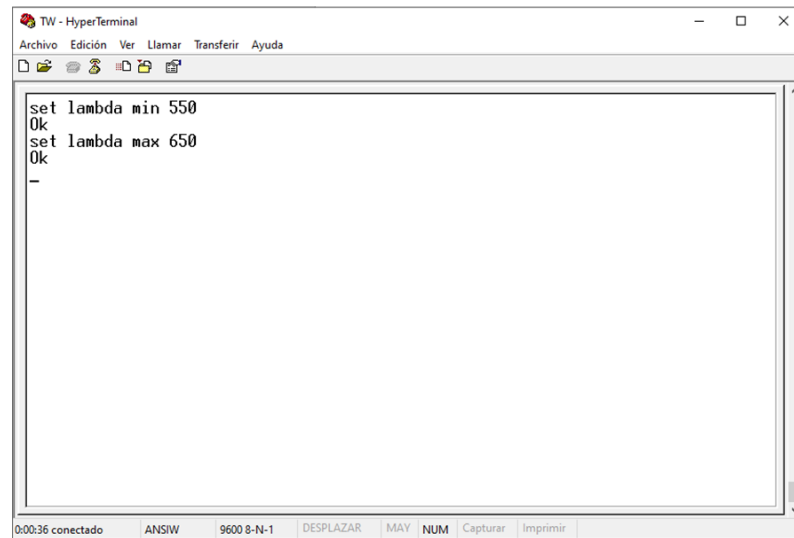
Bits de parada: 1

Control de flujo: Ninguno

6. At this point we should be connected to the BOREAL. In case of not visualizing the introduced commands go to: File/Proprieties/Configuration/ASCII Configuration and mark the case near Eco of characters.

The left screenshot shows the 'Configuración' tab in the software interface. The 'Configuración ASCII...' button is highlighted with a red box. The right screenshot shows the 'Configuración ASCII' dialog box. The 'Eco de los caracteres escritos localmente' checkbox is checked and highlighted with a red box.

7. Now, the introduced commands should be visualized. Next, write a command and press your keyboard Enter. Example: <set lambda min 550><enter>;<set lambda max 650><enter>. If the introduced command is accepted the BOREAL will answer with “Ok” via serial communication.



```

TW - HyperTerminal
Archivo Edición Ver Llamar Transferir Ayuda
set lambda min 550
Ok
set lambda max 650
Ok
-
0:00:36 conectado ANSIW 9600 8-N-1 DESPLAZAR MAY NUM Capturar Imprimir

```

Commands List

Command	Parameter	Description
Set lambda min xxx	Wavelength in nm	Sets the shortest wavelength of the bandwidth to the indicated value. The value formed by three characters must be within the allowed range, depending on BOREAL version.
Set lambda max xxx	Wavelength in nm	Sets the longest wavelength of the bandwidth to the indicated value. The value formed by three characters must be within the allowed range, depending on BOREAL version.
Set lambda xxxxxx	Wavelengths in nm	Sets at the same time the longest and shortest wavelengths of the bandwidth to the indicated values. The value, formed by six characters, must indicate firstly the longest and then the shortest. Example: 620550
move m1 xxxxx	Motor 1 position in steps	Sets the position of the motor 1 (that changes minimum wavelength) to the indicated value. The value must be within the allowed range depending on your BOREAL.

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move m2 xxxxx	Motor 2 position in steps	Sets the position of the motor 2 (that changes maximum wavelength) to the indicated value. The value must be within the allowed range depending on your BOREAL.
ps1	-	Shows general information (list below this table)
Home	-	Activates the system to change from the actual bandwidth to the default bandwidth (initial position when turning the system on)

PS1 returns:

- 001: Device status.
- 002: Motor 1 position.
- 003: Motor 2 position.
- 004: Motor 3 position.
- 005: Lambda max.
- 006: Lambda min.
- 007: Interlock status.
- 008: Shutter status
- 009: HOME position Motor 1
- 010: HOME position Motor 2
- 011: Monitor error

Operation

WARNING

THE LASER RADIATION AT THE OUTPUT OF THIS UNIT MAY BE HARMFUL. PLEASE FOLLOW ALL THE SAFETY INSTRUCTIONS INDICATED IN THE “USER SAFETY” SECTION BEFORE OPERATING THE LASER AND BOREAL LASER ACCESSORY.

After achieving all the electrical connections according to the previous instructions, please follow the next steps to operate the BOREAL laser accessory:

1. Make sure that the supercontinuum laser output collimator was properly connected to the BOREAL laser accessory input as indicated in the section “Optical Connections”.
2. When the BOREAL shows “READY” in the UI, start the Supercontinuum emission as indicated in the laser user manual.
3. Select the desired band changing the $\lambda_{\min}$ and $\lambda_{\max}$ parameters or Central λ and Bandwidth parameters.
4. Switch ON/OFF the BOREAL 's Shutter in case you've purchased a version including this with this option.

At this point you must be able to see the selected band light at the BOREAL laser accessory output.

WARNINGS AND FAILURE TO FUNCTION

WARNING

THE LASER RADIATION AT THE OUTPUT OF THIS LASER ACCESSORY MAY BE HARMFUL. PLEASE FOLLOW ALL THE SAFETY INSTRUCTIONS FROM “USER SAFETY” SECTION BEFORE OPERATING THE LASER.

IMPORTANT

BOREAL UNIT IS A LASER ACCESSORY AND MUST NOT BE USED AS A STAND-ALONE DEVICE. IT MAY ONLY BE OPERATED INSIDE A COMPLETE SYSTEM OR SETUP PROVIDING ALL MANDATORY SAFETY MECHANISMS

WARNING

WHEN USING YOUR ICEBLINK SERIES LASER, AVOID OPTICAL BACK REFLECTIONS TO THE SYSTEM. THE SYSTEM IS PROTECTED AGAINST INCIDENTAL LOW POWER BACK REFLECTIONS. OPTICAL BACK REFLECTIONS OF > 50 MW AVG. POWER MAY ALTER THE CORRECT OPERATION OF THE LASER, AND EVEN DAMAGE IT IRREVERSIBLY. TO AVOID THIS, SIMPLY TILT SLIGHTLY REFLECTIVE OPTICAL COMPONENTS IN YOUR SETUPS SO THAT DIRECT BACK REFLECTIONS ARE ELIMINATED.

WARNING

DO NOT FORCE THE FIBER TERMINATION ASSEMBLY.

WARNING

THE INTERLOCK CONNECTION MUST SUPPORT 24V/10A.

WARNING_____

AVOID LOCATIONS WHERE THE LASER IS EXPOSED TO EXTREME TEMPERATURES AND HIGH HUMIDITY.

WARNING_____

AVOID LOCATIONS WHERE THE LASER IS EXPOSED TO MECHANICAL VIBRATIONS.

WARNING_____

ANY KIND OF OPERATION OF MANTAINANCE MUST BE PERFORMED BY OUR QUALIFIED LASER TECHNICIANS OR UNDER THEIR GUIDANCE WHEN THEY SPECIFICALLY INDICATE SO.

WARNING_____

IN CASE OF LASER MALFUNCTION, FOR YOUR SELF-PROTECTION, DISCONNECT THE INTERLOCK CONNECTOR.

CUSTOMER SERVICE

Technical Support

Information and advice about the operation of any FYLA product are available from our technical support engineers. For the quickest response, ask for “Technical Support” at support@fyla.com, votgon@fyla.com and ogarcia@fyla.com including the model and serial number of your product.

Hours: 9:00 to 13:00 and 15:00 to 18:00, Monday to Thursday, Friday 9:00 to 14:30 GMT +1 (excluding holidays).

Phone: (+34) 607971021

for e-mail inquiries, we typically respond within one business day.

Service

In the event that your device malfunctions or becomes damaged, please contact FYLA for a return authorization number and instructions on shipping the unit back for evaluation and repair/replacement.

FYLA Laser S.L.
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